

Mastering Quantum Mechanics — Problem 9.7

Harmonic oscillators oscillating!

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Problem 9.7

Harmonic oscillators oscillating! At $t = 0$, $\Psi(x, 0) = \frac{1}{\sqrt{2}}(\varphi_0(x) + \varphi_1(x))$ with φ_0, φ_1 real normalized eigenstates ($n = 0, 1$). (1) Write $\Psi(x, t)$ and $|\Psi(x, t)|^2$. (2) Find $\langle \hat{x} \rangle(t)$. (3) Find $\langle \hat{p} \rangle(t)$. (4) For an *arbitrary* oscillator state, find the smallest T with $|\Psi(x, t)|^2 = |\Psi(x, t + T)|^2$, and write $\Psi(x, t + T)$ in terms of $\Psi(x, t)$.

Hints & Orientation

- (1) Each eigenstate evolves by its phase $e^{-iE_n t/\hbar}$, $E_n = \hbar\omega(n + \frac{1}{2})$. Factor out the common $e^{-i\omega t/2}$; the cross term in $|\Psi|^2$ oscillates at the *difference* frequency $(E_1 - E_0)/\hbar = \omega$.
- (2)–(3) Only off-diagonal matrix elements survive ($\langle \varphi_n | \hat{x} | \varphi_n \rangle = 0$). Use $\hat{x}, \hat{p}$ in ladder operators (Eq. (9.3.12)) and (9.4.26): only $\langle \varphi_0 | \hat{x} | \varphi_1 \rangle$ etc. are nonzero. Expect $\langle \hat{p} \rangle = m \frac{d}{dt} \langle \hat{x} \rangle$.
- (4) $|\Psi|^2$ is unchanged iff $\Psi(t + T) = e^{i\theta} \Psi(t)$ (a global phase). That needs $e^{-iE_n T/\hbar}$ to be n -independent up to a common phase $\Rightarrow \omega T = 2\pi$.

Solution

(1) Time evolution

With $E_0 = \frac{1}{2}\hbar\omega$, $E_1 = \frac{3}{2}\hbar\omega$,

$$\Psi(x, t) = \frac{1}{\sqrt{2}} \left(\varphi_0 e^{-i\omega t/2} + \varphi_1 e^{-3i\omega t/2} \right) = \frac{e^{-i\omega t/2}}{\sqrt{2}} \left(\varphi_0 + \varphi_1 e^{-i\omega t} \right). \quad (1)$$

Since φ_0, φ_1 are real,

$$|\Psi(x, t)|^2 = \frac{1}{2} \left(\varphi_0^2 + \varphi_1^2 + 2\varphi_0\varphi_1 \cos \omega t \right). \quad (2)$$

(2) $\langle \hat{x} \rangle(t)$

The only nonzero element is $\langle \varphi_0 | \hat{x} | \varphi_1 \rangle = \langle \varphi_1 | \hat{x} | \varphi_0 \rangle = \frac{L_0}{\sqrt{2}}$ (from Eqs. (9.3.12), (9.4.26)), so

$$\langle \hat{x} \rangle(t) = \frac{1}{2} \cdot \frac{L_0}{\sqrt{2}} (e^{-i\omega t} + e^{i\omega t}) = \frac{L_0}{\sqrt{2}} \cos \omega t = \sqrt{\frac{\hbar}{2m\omega}} \cos \omega t. \quad (3)$$

(3) $\langle \hat{p} \rangle(t)$

With $\langle \varphi_1 | \hat{p} | \varphi_0 \rangle = \frac{i\hbar}{\sqrt{2}L_0}$ and $\langle \varphi_0 | \hat{p} | \varphi_1 \rangle = -\frac{i\hbar}{\sqrt{2}L_0}$,

$$\langle \hat{p} \rangle(t) = \frac{1}{2} \cdot \frac{i\hbar}{\sqrt{2}L_0} (e^{i\omega t} - e^{-i\omega t}) = \boxed{-\frac{\hbar}{\sqrt{2}L_0} \sin \omega t = -\sqrt{\frac{m\omega\hbar}{2}} \sin \omega t}. \quad (4)$$

Consistency: $m \frac{d}{dt} \langle \hat{x} \rangle = -m\omega \frac{L_0}{\sqrt{2}} \sin \omega t = -\sqrt{\frac{m\omega\hbar}{2}} \sin \omega t = \langle \hat{p} \rangle$. ✓

(4) Period of $|\Psi|^2$

For a general state $\Psi(x, t) = \sum_n c_n e^{-iE_n t/\hbar} \varphi_n$,

$$\Psi(x, t+T) = \sum_n c_n e^{-iE_n t/\hbar} e^{-iE_n T/\hbar} \varphi_n. \quad (5)$$

$|\Psi|^2$ is unchanged iff the extra phases are common to all n , i.e. $e^{-iE_n T/\hbar} = e^{-i\omega(n+1/2)T}$ is n -independent up to a global factor. This requires $e^{-i\omega n T} = 1$ for all integers n , i.e. $\omega T = 2\pi$:

$$\boxed{T = \frac{2\pi}{\omega}} \quad (\text{the classical oscillation period}). \quad (6)$$

At this T the residual common phase is $e^{-i\omega T/2} = e^{-i\pi} = -1$, hence

$$\boxed{\Psi(x, t+T) = -\Psi(x, t)}. \quad (7)$$

So the probability density is periodic with $T = 2\pi/\omega$, while the wave function itself returns only after $2T = 4\pi/\omega$ (the extra sign is the zero-point phase $e^{-i\omega T/2}$).

Checks

- $\langle \hat{x} \rangle$, $\langle \hat{p} \rangle$ trace the classical ellipse and oscillate at ω — Ehrenfest's theorem for the oscillator.
- Amplitudes obey $\langle \hat{p} \rangle_{\max} = m\omega \langle \hat{x} \rangle_{\max}$, as for a classical oscillator.
- $|\Psi|^2$ depends on t only through $\cos \omega t$, manifestly of period $2\pi/\omega$, matching part (4).